

ZHONGSHAN, AntarcticaWMO: **895730**Lat: **69.374S**Lon: **76.372E**Elev: **18**StdP: **101.11**Time Zone: **5.00 (E05)**Period: **91-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -34.3	(c) -31.0	(d) -38.6	(e) 0.1	(f) -32.4	(g) -36.6	(h) 0.1	(i) -29.4	(j) 30.8	(k) -3.0	(l) 27.4	(m) -9.6	(n) 2.0	(o) 250	(p) 1.244

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 3.5	(c) 4.5	(d) 0.5	(e) 3.6	(f) -0.1	(g) 2.8	(h) -0.6	(i) 0.9	(j) 3.6	(k) 0.3	(l) 2.9	(m) -0.3	(n) 2.3	(o) 3.9	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
-1.1	3.4	0.7	-2.3	3.1	0.5	-3.3	2.9	0.4	11.3	3.8	10.2	3.1	9.2	2.4	3.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	20.1	17.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			-36.7	1.8	3.7	1.1	-39.3	2.5	-41.5	3.2	-43.6	3.8	-46.3	4.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-10.4	0.4	-3.1	-8.8	-13.8	-16.1	-15.2	-17.6	-16.4	-16.0	-12.4	-5.0	-0.1	(6)
(7)		DBStd	8.01	1.57	2.38	3.67	4.80	5.81	5.66	7.37	6.11	6.05	4.19	3.43	1.95	(7)
(8)		HDD10.0	7431	298	367	582	713	808	755	856	817	779	693	449	314	(8)
(9)		HDD18.3	10473	557	600	841	963	1067	1005	1114	1075	1029	952	699	572	(9)
(10)		CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.4	5.3	7.9	8.5	7.4	8.0	8.6	7.9	8.5	7.5	7.5	6.4	5.9	(14)
(15)	Precipitation	PrecAvg	336	38	39	32	21	23	24	27	24	21	22	34	31	(15)
(16)		PrecMax	450	107	72	69	54	53	56	74	80	53	52	85	61	(16)
(17)		PrecMin	170	12	19	11	8	8	6	1	3	1	6	12	9	(17)
(18)		PrecStd	52	19	15	14	11	11	10	15	15	11	10	17	13	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	3.8	-0.7	-2.9	-2.4	-2.8	-2.5	-3.5	-1.9	3.0	5.9	(19)	
(20)			MCWB	1.3	0.0	-3.2	-4.8	-4.8	-5.1	-4.2	-5.1	-5.2	-4.7	-1.1	1.5	(20)
(21)		2%	DB	4.5	1.7	-1.9	-4.5	-4.4	-4.8	-4.3	-5.4	-4.5	-3.5	1.7	4.6	(21)
(22)			MCWB	0.5	-1.2	-4.0	-5.9	-6.0	-6.5	-5.3	-7.1	-6.4	-5.6	-1.6	0.5	(22)
(23)		5%	DB	3.7	1.0	-2.9	-5.6	-5.8	-5.9	-6.0	-6.8	-6.4	-4.7	0.7	3.8	(23)
(24)			MCWB	0.0	-1.7	-4.7	-7.1	-7.1	-7.5	-7.0	-8.3	-8.2	-6.6	-2.2	0.0	(24)
(25)		10%	DB	2.9	0.2	-4.0	-7.2	-8.1	-7.3	-7.8	-8.2	-8.0	-6.1	-0.2	3.0	(25)
(26)			MCWB	-0.4	-2.4	-5.7	-8.5	-9.6	-8.7	-8.9	-9.5	-9.5	-7.9	-2.8	-0.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	0.7	-2.6	-4.4	-4.5	-4.6	-3.8	-5.1	-4.7	-3.7	-0.4	1.9	(27)
(28)			MCD B	4.2	2.3	-1.3	-3.3	-2.8	-2.9	-3.1	-3.0	-3.8	-2.7	1.7	5.5	(28)
(29)		2%	WB	1.1	-0.5	-3.7	-5.8	-5.9	-6.3	-5.2	-6.9	-6.4	-5.3	-1.3	0.8	(29)
(30)			MCD B	3.7	1.1	-2.3	-4.5	-4.8	-5.2	-4.5	-5.7	-4.6	-3.7	1.1	3.8	(30)
(31)		5%	WB	0.5	-1.3	-4.5	-7.2	-7.2	-7.3	-6.8	-8.2	-8.1	-6.5	-1.8	0.2	(31)
(32)			MCD B	3.0	0.3	-3.0	-6.0	-5.8	-6.0	-5.9	-6.9	-6.5	-4.8	0.5	3.2	(32)
(33)		10%	WB	-0.1	-2.0	-5.6	-8.5	-9.5	-8.7	-8.9	-9.4	-9.4	-7.8	-2.6	-0.3	(33)
(34)			MCD B	2.4	-0.1	-4.0	-7.2	-8.1	-7.3	-7.8	-8.2	-8.1	-6.1	-0.2	2.5	(34)
(35)	Mean Daily Temperature Range	MDBR	3.5	3.2	3.4	4.1	4.5	4.7	4.9	5.4	4.6	4.8	4.9	4.3	(35)	
(36)		5% DB	MCD BR	4.6	4.1	3.6	4.6	5.0	4.8	3.9	5.3	3.7	5.0	5.1	5.3	(36)
(37)			MCWBR	3.3	3.4	3.3	4.6	4.9	4.6	3.8	4.8	3.3	4.5	4.0	3.8	(37)
(38)			5% WB	MCD BR	4.1	3.3	3.4	4.9	4.7	5.3	3.9	5.6	4.4	5.2	4.3	4.7
(39)		MCWBR		3.1	3.2	3.3	5.0	4.7	5.2	3.8	5.3	4.2	4.8	3.6	3.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.246	0.219	0.163	0.121	N/A	0.124	0.158	0.202	0.217	0.242	0.258	(40)	
(41)		taud	2.317	2.347	2.308	2.104	1.904	N/A	1.884	2.061	2.222	2.236	2.247	2.264	(41)	
(42)		Ebn at Noon	992	943	858	650	91	0	59	636	864	971	1004	1006	(42)	
(43)		Edh at Noon	106	88	66	34	3	0	3	35	69	96	113	116	(43)	
(44)	All-Sky Solar Radiation	RadAvg	8.03	5.22	2.56	0.75	0.06	0.00	0.01	0.38	1.87	4.51	7.41	9.07	(44)	
(45)		RadStd	0.29	0.25	0.12	0.04	0.00	0.00	0.00	0.02	0.10	0.17	0.25	0.32	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page